In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [3]:
```
dataframe = pd.read_csv("Zomato-data-.csv")
print(dataframe.head())
```

```
                    name online_order book_table   rate  votes  \
0                  Jalsa          Yes        Yes  4.1/5    775
1         Spice Elephant          Yes         No  4.1/5    787
2        San Churro Cafe          Yes         No  3.8/5    918
3  Addhuri Udupi Bhojana           No         No  3.7/5     88
4          Grand Village           No         No  3.8/5    166

   approx_cost(for two people) listed_in(type)
0                          800          Buffet
1                          800          Buffet
2                          800          Buffet
3                          300          Buffet
4                          600          Buffet
```

In [4]:
```
def handleRate(value):
    value=str(value).split('/')
    value=value[0];
    return float(value)

dataframe['rate']=dataframe['rate'].apply(handleRate)
print(dataframe.head())
```

```
                    name online_order book_table  rate  votes  \
0                  Jalsa          Yes        Yes   4.1    775
1         Spice Elephant          Yes         No   4.1    787
2        San Churro Cafe          Yes         No   3.8    918
3  Addhuri Udupi Bhojana           No         No   3.7     88
4          Grand Village           No         No   3.8    166

   approx_cost(for two people) listed_in(type)
0                          800          Buffet
1                          800          Buffet
2                          800          Buffet
3                          300          Buffet
4                          600          Buffet
```

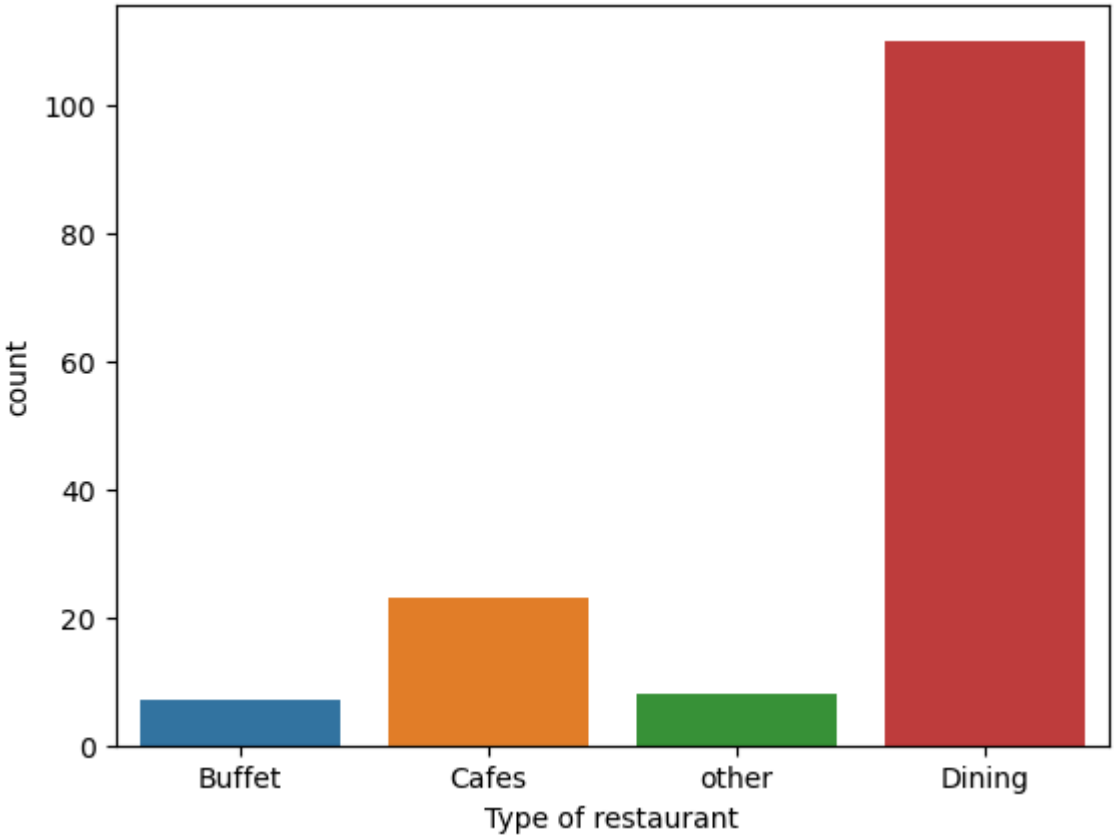
In [7]:
```
print(dataframe.isnull().sum())
```

```
name                           0
online_order                   0
book_table                     0
rate                           0
votes                          0
approx_cost(for two people)    0
listed_in(type)                0
dtype: int64
```

1) Listed_in (type) identify column popular restaurant categories.

In [8]:
```
sns.countplot(x=dataframe['listed_in(type)'])
plt.xlabel("Type of restaurant")
```

Out[8]:
```
Text(0.5, 0, 'Type of restaurant')
```



2) votes by Restaurant type

In [10]:
```
grouped_data = dataframe.groupby('listed_in(type)')['votes'].sum()
result = pd.DataFrame({'votes': grouped_data})
plt.plot(result, c='green', marker='o')
plt.xlabel('Type of restaurant')
plt.ylabel('Votes')
```

Out[10]:
```
Text(0, 0.5, 'Votes')
```

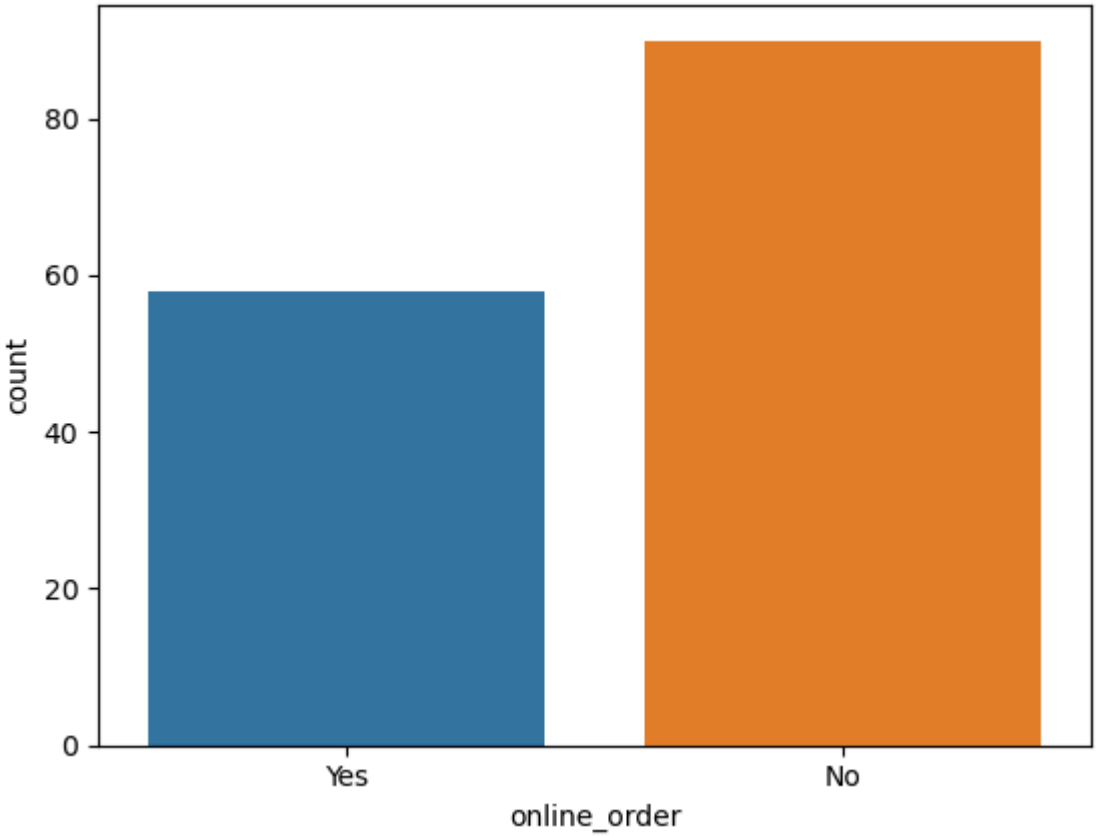
In [11]:
```
max_votes = dataframe['votes'].max()
restaurant_with_max_votes = dataframe.loc[dataframe['votes'] == max_votes, 'name']

print('Restaurant(s) with the maximum votes:')
print(restaurant_with_max_votes)
```

```
Restaurant(s) with the maximum votes:
38    Empire Restaurant
Name: name, dtype: object
```

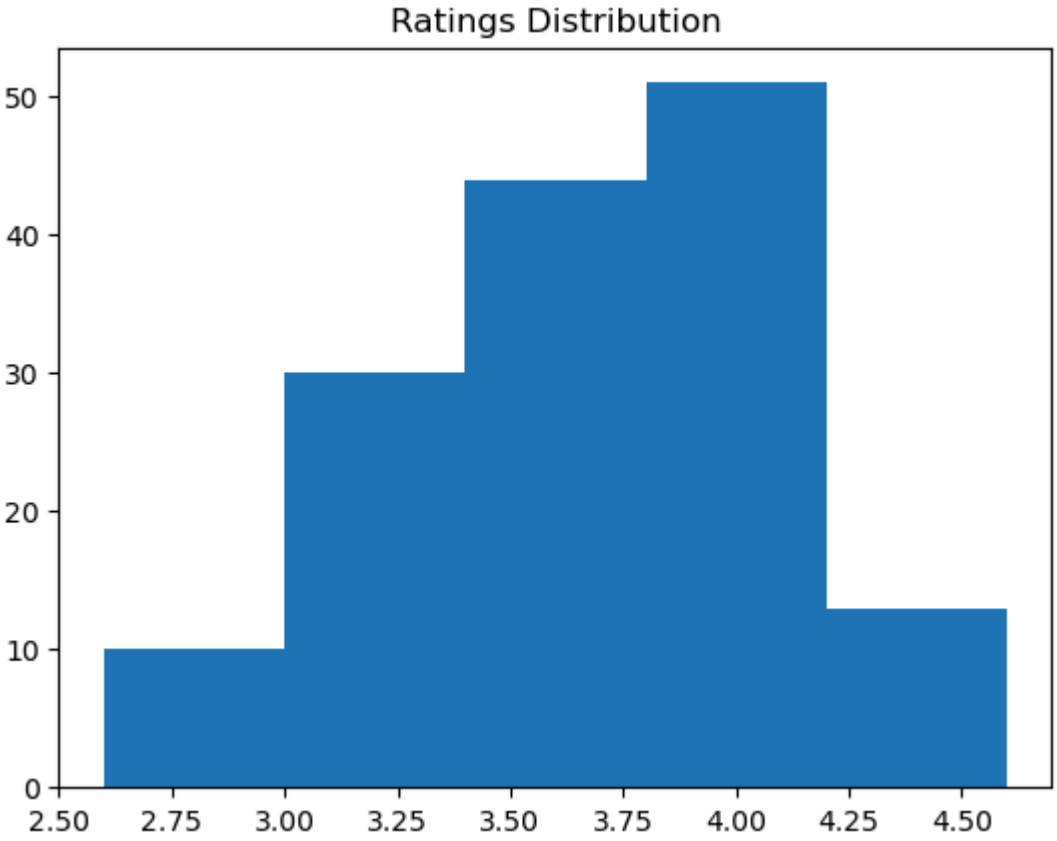
In [12]:
```
sns.countplot(x=dataframe['online_order'])
```

Out[12]:
```
<AxesSubplot:xlabel='online_order', ylabel='count'>
```



Analyze Ratings

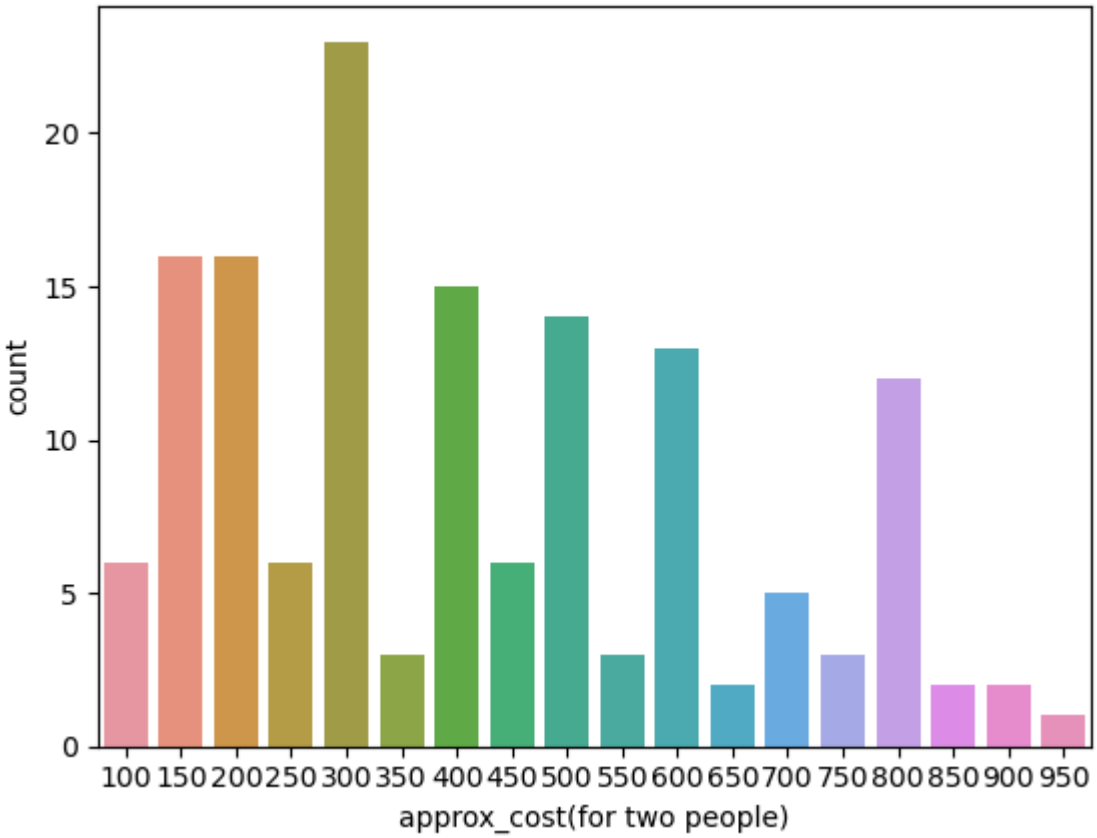
In [13]:
```
plt.hist(dataframe['rate'],bins=5)
plt.title('Ratings Distribution')
plt.show()
```



Approx_cost(for two people)

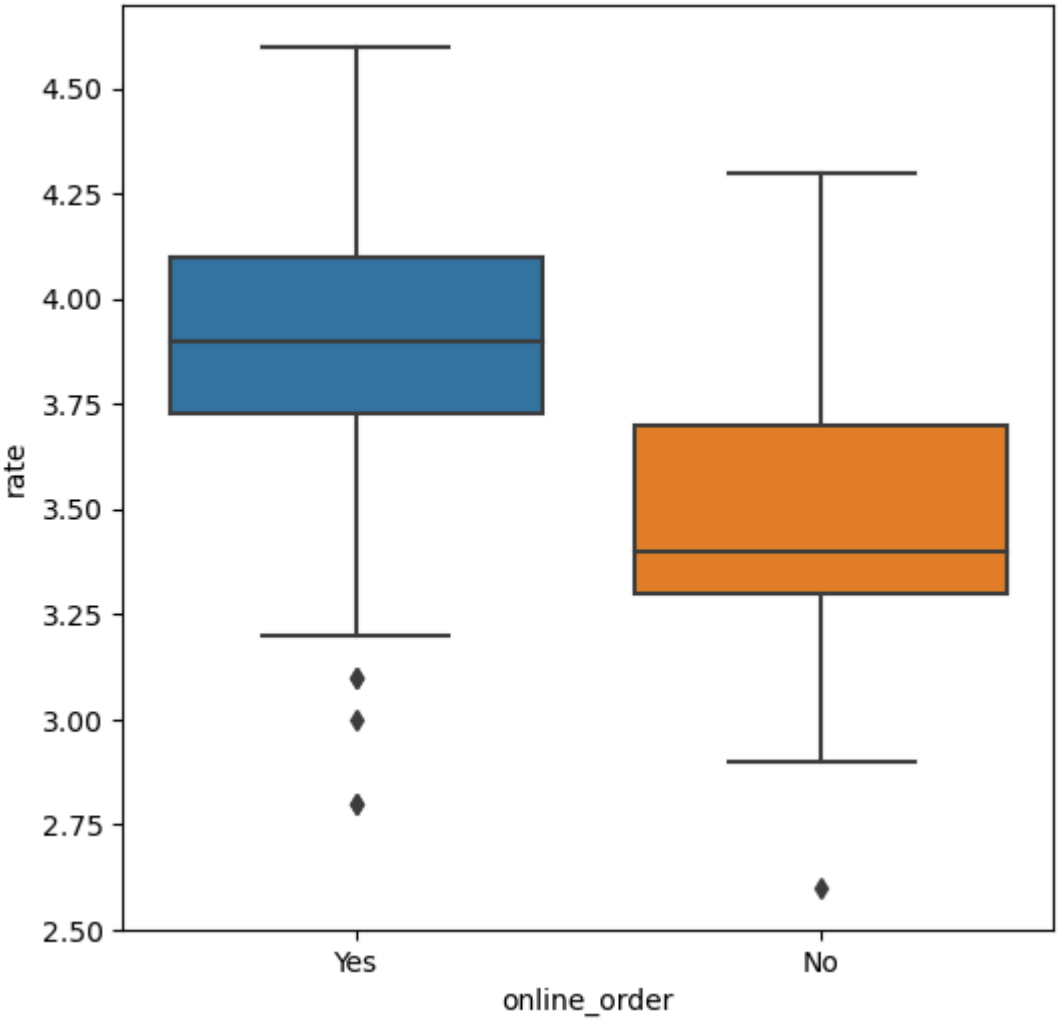
In [14]:
```
couple_data=dataframe['approx_cost(for two people)']
sns.countplot(x=couple_data)
```

Out[14]:
```
<AxesSubplot:xlabel='approx_cost(for two people)', ylabel='count'>
```



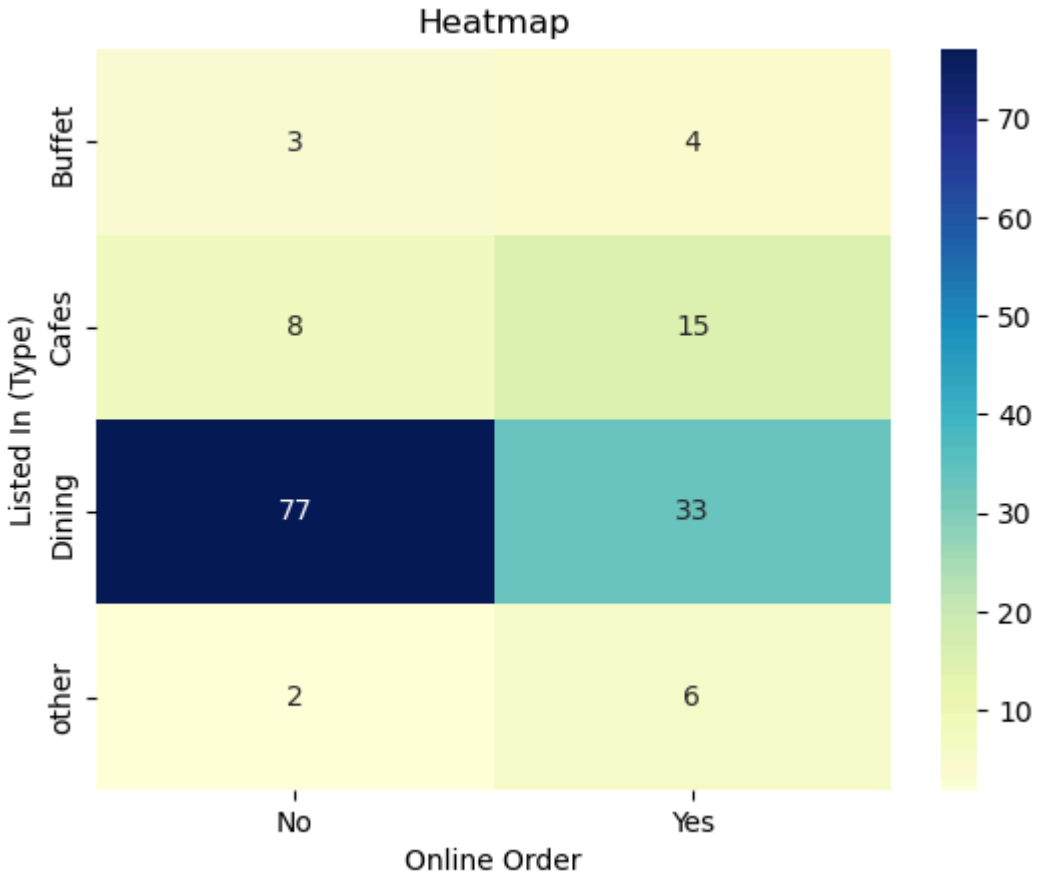
In [15]:
```
plt.figure(figsize = (6,6))
sns.boxplot(x = 'online_order', y = 'rate', data = dataframe)
```

Out[15]:
```
<AxesSubplot:xlabel='online_order', ylabel='rate'>
```



Relationship between order mode (online_order) and restaurant type (listed_in(type))

In [16]:
```
pivot_table = dataframe.pivot_table(index='listed_in(type)', columns='online_order'
sns.heatmap(pivot_table, annot=True, cmap='YlGnBu', fmt='d')
plt.title('Heatmap')
plt.xlabel('Online Order')
plt.ylabel('Listed In (Type)')
plt.show()
```


Heatmap
Listed In (Type)
Buffet
Cafes
Dining
other
3
4
8
15
77
33
2
6
No
Yes
Online Order
70
60
50
40
30
20
10

In [ ]: